Eleventh Grade Tile Task

Standard A.APR.2

Name: _____ Date: _____

11

1. a) Remainder of $(x^2 + 4x + 4) \div (x - 1)$ ☐

b) Remainder of $(x^2 + 2x + 0) \div (x - 1)$ ☐

c) Remainder of $(x^2 + 1x + 3) \div (x - 1)$ ☐

2. a) Remainder of $(x^2 + 0x + 1) \div (x - 1)$ ☐

b) Remainder of $(x^2 + 3x + 3) \div (x - 1)$ ☐

c) Remainder of $(x^2 + 1x + 4) \div (x - 1)$ ☐

3. a) Remainder of $(x^2 + 0x + 0) \div (x - 1)$ ☐

b) Remainder of $(x^2 + 3x + 4) \div (x - 1)$ ☐

c) Remainder of $(x^2 + 2x + 1) \div (x - 1)$ ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eleventh Grade Tile Task

Standard A.REI.2

Name: _____ Date: _____

11

1. **a)** Solve x/4 = 2: x = ? ☐

b) Solve ✓x = 2: x = ? ☐

c) Solve x/9 = 1: x = ? ☐

2. **a)** Solve x/3 = 1: x = ? ☐

b) Solve x/5 = 1: x = ? ☐

c) Solve ✓x = 0: x = ? ☐

3. **a)** Solve ✓x = 1: x = ? ☐

b) Solve x/2 = 1: x = ? ☐

c) Solve x/7 = 1: x = ? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard A.REI.7

Name: _____ Date: _____

11

1. a) Solve x + y = 7, x = 4: y = ? ☐

b) Solve x + y = 3, x = 2: y = ? ☐

c) Solve x + y = 9, x = 0: y = ? ☐

2. a) Solve x + y = 3, x = 1: y = ? ☐

b) Solve x + y = 9, x = 3: y = ? ☐

c) Solve x + y = 8, x = 0: y = ? ☐

3. a) Solve x + y = 5, x = 0: y = ? ☐

b) Solve x + y = 6, x = 2: y = ? ☐

c) Solve x + y = 6, x = 6: y = ? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard A.REI.11

Name: _____ Date: _____

11

1. **a)** x-coordinate where y = 2x meets y = 4 ☐

b) x-coordinate where y = 1x meets y = 9 ☐

c) x-coordinate where y = 1x meets y = 8 ☐

2. **a)** x-coordinate where y = 1x meets y = 6 ☐

b) x-coordinate where y = 3x meets y = 0 ☐

c) x-coordinate where y = 2x meets y = 6 ☐

3. **a)** x-coordinate where y = 3x meets y = 3 ☐

b) x-coordinate where y = 1x meets y = 5 ☐

c) x-coordinate where y = 2x meets y = 8 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard A.SSE.3

Name: _____ Date: _____

11

1. a) Smaller root of $x^2 - 17x + 72$ ☐

b) Smaller root of $x^2 - 9x + 18$ ☐

c) Smaller root of $x^2 - 15x + 56$ ☐

2. a) Smaller root of $x^2 - 11x + 18$ ☐

b) Smaller root of $x^2 - 3x + 2$ ☐

c) Smaller root of $x^2 - 4x + 0$ ☐

3. a) Smaller root of $x^2 - 13x + 40$ ☐

b) Smaller root of $x^2 - 12x + 32$ ☐

c) Smaller root of $x^2 - 14x + 48$ ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eleventh Grade Tile Task

Standard F.BF.3

Name: _____ Date: _____

11

1. a) $f(x) = x^2 + 4$: value at $x = 0$ ☐

b) $f(x) = x^2 + 0$: value at $x = 0$ ☐

c) $f(x) = x^2 + 2$: value at $x = 0$ ☐

2. a) $f(x) = x^2 + 3$: value at $x = 0$ ☐

b) $f(x) = x^2 + 6$: value at $x = 0$ ☐

c) $f(x) = x^2 + 9$: value at $x = 0$ ☐

3. a) $f(x) = x^2 + 5$: value at $x = 0$ ☐

b) $f(x) = x^2 + 1$: value at $x = 0$ ☐

c) $f(x) = x^2 + 7$: value at $x = 0$ ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.BF.4

Name: _____ Date: _____

11

1. a) $f(x) = x + 1$. Find $f^1(8)$. ☐

b) $f(x) = x + 4$. Find $f^1(6)$. ☐

c) $f(x) = x + 3$. Find $f^1(3)$. ☐

2. a) $f(x) = x + 1$. Find $f^1(6)$. ☐

b) $f(x) = x + 0$. Find $f^1(8)$. ☐

c) $f(x) = x + 0$. Find $f^1(9)$. ☐

3. a) $f(x) = x + 1$. Find $f^1(5)$. ☐

b) $f(x) = x + 0$. Find $f^1(6)$. ☐

c) $f(x) = x + 5$. Find $f^1(8)$. ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.IF.6

Name: _____ Date: _____

11

1. **a)** Average rate of change of $y = x^2$ from $x = 0$ to $x = 4$ ☐

b) Average rate of change of $y = x^2$ from $x = 0$ to $x = 1$ ☐

c) Average rate of change of $y = x^2$ from $x = 0$ to $x = 2$ ☐

2. **a)** Average rate of change of $y = x^2$ from $x = 0$ to $x = 6$ ☐

b) Average rate of change of $y = x^2$ from $x = 0$ to $x = 9$ ☐

c) Average rate of change of $y = x^2$ from $x = 0$ to $x = 7$ ☐

3. **a)** Average rate of change of $y = x^2$ from $x = 0$ to $x = 8$ ☐

b) Average rate of change of $y = x^2$ from $x = 0$ to $x = 5$ ☐

c) Average rate of change of $y = x^2$ from $x = 0$ to $x = 3$ ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eleventh Grade Tile Task

Standard F.IF.7

Name: _____ Date: _____

11

1. a) y-intercept of $y = x^2 + 0$ ☐

b) y-intercept of $y = x^2 + 8$ ☐

c) y-intercept of $y = x^2 + 4$ ☐

2. a) y-intercept of $y = x^2 + 5$ ☐

b) y-intercept of $y = x^2 + 2$ ☐

c) y-intercept of $y = x^2 + 9$ ☐

3. a) y-intercept of $y = x^2 + 1$ ☐

b) y-intercept of $y = x^2 + 6$ ☐

c) y-intercept of $y = x^2 + 7$ ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.IF.8

Name: _____ Date: _____

11

1. a) f(x) = x + 6: f(1) ☐

b) f(x) = x + 4: f(1) ☐

c) y = 2^x: value at x = 3 ☐

2. a) f(x) = x + 3: f(1) ☐

b) f(x) = x + 8: f(1) ☐

c) y = 2^x: value at x = 1 ☐

3. a) f(x) = x + 5: f(1) ☐

b) y = 3^x: value at x = 1 ☐

c) y = 3^x: value at x = 0 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.IF.9

Name: _____ Date: _____

11

1. a) Slope of f(x) = 2x + 1 ☐

b) Slope of f(x) = 1x + 0 ☐

c) Slope of f(x) = 9x + 0 ☐

2. a) Slope of f(x) = 7x + 0 ☐

b) Slope of f(x) = 8x + 1 ☐

c) Slope of f(x) = 0x + 0 ☐

3. a) Slope of f(x) = 6x + 1 ☐

b) Slope of f(x) = 5x + 0 ☐

c) Slope of f(x) = 3x + 0 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.LE.4

Name: _____ Date: _____

11

1. **a)** log base 10 of 10^2 ☐
   b) log base 10 of 10^5 ☐
   c) log base 10 of 10^9 ☐

2. **a)** log base 10 of 10^1 ☐
   b) log base 10 of 10^4 ☐
   c) log base 10 of 10^8 ☐

3. **a)** log base 10 of 10^6 ☐
   b) log base 10 of 10^3 ☐
   c) log base 10 of 10^7 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.TF.1

Name: _____ Date: _____

11

1. **a)** How many full turns in 2π radians? ☐

b) How many full turns in 4π radians? ☐

c) How many full turns in 6π radians? ☐

2. **a)** How many full turns in 0π radians? ☐

b) How many full turns in 18π radians? ☐

c) How many full turns in 10π radians? ☐

3. **a)** How many full turns in 12π radians? ☐

b) How many full turns in 8π radians? ☐

c) How many full turns in 16π radians? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard F.TF.2

Name: _____ Date: _____

11

1. **a)** Points where the unit circle crosses the x-axis? ☐

b) Arc length on the unit circle for an angle of 1 radians ☐

c) Arc length on the unit circle for an angle of 9 radians ☐

2. **a)** Arc length on the unit circle for an angle of 0 radians ☐

b) Quadrants of the unit circle? ☐

c) Area of the unit circle ($\pi \approx 3$)? ☐

3. **a)** Arc length on the unit circle for an angle of 8 radians ☐

b) Arc length on the unit circle for an angle of 5 radians ☐

c) Arc length on the unit circle for an angle of 7 radians ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard F.TF.5

Name: _____ Date: _____

11

1. a) Amplitude of y = 3 sin x

b) Midline of y = 5 + sin x

c) Midline of y = 0 + sin x

2. a) Amplitude of y = 9 sin x

b) Amplitude of y = 6 sin x

c) Amplitude of y = 4 sin x

3. a) Midline of y = 1 + sin x

b) Amplitude of y = 8 sin x

c) Amplitude of y = 2 sin x

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard N.CN.1

Name: _____ Date: _____

11

1. **a)** Imaginary part of 1 + 2i ☐

b) Real part of 4 + 1i ☐

c) Real part of 3 + 0i ☐

2. **a)** Imaginary part of 1 + 9i ☐

b) Imaginary part of 0 + 8i ☐

c) Real part of 7 + 0i ☐

3. **a)** Real part of 5 + 0i ☐

b) Imaginary part of 1 + 1i ☐

c) Real part of 0 + 0i ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eleventh Grade Tile Task

Standard N.CN.2

Name: _____ Date: _____

11

1. **a)** Real part of (4 + 0i) + (2 + 0i) ☐

b) Real part of (3 + 0i) + (1 + 1i) ☐

c) Real part of (0 + 0i) + (0 + 0i) ☐

2. **a)** Real part of (4 + 1i) + (3 + 0i) ☐

b) Real part of (2 + 1i) + (3 + 0i) ☐

c) Real part of (1 + 0i) + (0 + 1i) ☐

3. **a)** Real part of (4 + 0i) + (4 + 0i) ☐

b) Real part of (3 + 1i) + (0 + 1i) ☐

c) Real part of (1 + 0i) + (1 + 1i) ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard N.CN.7

Name: _____ Date: _____

11

1. a) Real part of the roots of $(x - 5)^2 + 1 = 0$ ☐

b) Real part of the roots of $(x - 9)^2 + 1 = 0$ ☐

c) Real part of the roots of $x^2 + 1 = 0$ ($k = 1$) ☐

2. a) Real part of the roots of $(x - 1)^2 + 1 = 0$ ☐

b) Real part of the roots of $(x - 3)^2 + 1 = 0$ ☐

c) Real part of the roots of $(x - 4)^2 + 1 = 0$ ☐

3. a) Real part of the roots of $(x - 2)^2 + 1 = 0$ ☐

b) Real part of the roots of $(x - 7)^2 + 1 = 0$ ☐

c) Real part of the roots of $(x - 8)^2 + 1 = 0$ ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: ELEVENTH GRADE

Eleventh Grade Tile Task

Standard S.ID.6

Name: _____ Date: _____

11

1. a) Slope of the best-fit line y = 1x + 1 ☐

b) Slope of the best-fit line y = 9x + 0 ☐

c) Slope of the best-fit line y = 7x + 1 ☐

2. a) Slope of the best-fit line y = 8x + 1 ☐

b) Slope of the best-fit line y = 6x + 0 ☐

c) Slope of the best-fit line y = 5x + 0 ☐

3. a) Slope of the best-fit line y = 3x + 0 ☐

b) Slope of the best-fit line y = 2x + 0 ☐

c) Slope of the best-fit line y = 0x + 1 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eleventh Grade Tile Task

Standard S.ID.7

Name: _____ Date: _____

11

1. a) Interpret the slope: y = 7x + 0, slope? ☐

b) Interpret the slope: y = 2x + 0, slope? ☐

c) Interpret the slope: y = 0x + 1, slope? ☐

2. a) Interpret the slope: y = 8x + 2, slope? ☐

b) Interpret the slope: y = 4x + 0, slope? ☐

c) Interpret the slope: y = 5x + 2, slope? ☐

3. a) Interpret the slope: y = 9x + 0, slope? ☐

b) Interpret the slope: y = 3x + 2, slope? ☐

c) Interpret the slope: y = 6x + 2, slope? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9